

Seasons of Code'2026

Indian Institute of Technology Bombay

Project ID 31

Data Structures and Algorithms

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Mentors

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Endterm Report

1 Intro to the Project

This project was structured to learn the concepts of Data Structure and Algorithms such that we could apply the learnings in a very optimised way considering the complexities of code. Going into this project, I gained some skills to solve unseen problems confidently under time pressure.

The goal of this project was therefore not just to accumulate theoretical knowledge, but to build structured, pattern-based problem-solving ability that could be applied reliably to new problems.

2 What was the process doing this project?

The following structured approach was followed throughout the project:

- Studied topics sequentially using Striver's A2Z DSA Course (take U forward Youtube channel), which provided structured theory along with video explanations for each concept.
- Mentors released a curated Google Sheet every two days, containing 9–10 questions per topic on leetcode, allowing each concept to be applied immediately after it was learnt.
- Maintained personal notes in VS Code, creating a separate file for every topic covered – consolidating key concepts, patterns, and edge cases for quick revision.
- Followed a consistent learning loop: learn a topic → solve the mentor-assigned question set → reinforce understanding through notes → move to the next topic.

This cycle ensured that every topic was reinforced through active practice rather than passive learning, and that a personal reference was built up alongside the course for future revision. I also participated in some Competitive Programming contests on CodeForces.

3 Results

By the end of the project, the following outcomes were achieved:

- Completed a full sweep of 9 major DSA topic areas, from arrays and traversal techniques through advanced graph algorithms.
- Solved the complete set of mentor-assigned questions across all problem sheets released over the course of the project.
- Built a personal, topic-wise notes repository in VS Code, serving as a structured revision resource covering every topic studied.
- Noticeable improvement in pattern recognition and problem-solving speed – topics that initially required significant time to approach (e.g. dynamic programming problems) became progressively faster to identify and solve as the project progressed.

4 Topics Covered

Over the course of this project, the following topics were studied in sequence – following structured video lectures, and then reinforced through mentor-assigned problem sets released every two days.

4.1 Arrays and Traversal

- Array traversal
- Prefix sums / prefix array technique
- Frequency counting
- Basic simulation

4.2 Two Pointers and Sliding Window

- Sliding window technique
- Two pointers

4.3 Searching and Sorting

- Binary search
- Sorting algorithms

4.4 Stacks and Queues

- Stacks
- Queues
- Deque
- Monotonic stack

4.5 Recursion and Backtracking

- Recursion
- Backtracking

4.6 Trees

- Binary trees
- Binary search trees (BST)

4.7 Greedy Algorithms and Heaps

- Greedy algorithms
- Heaps

4.8 Dynamic Programming

- DP fundamentals
- DP on subsequences
- DP on strings
- DP on stocks

4.9 Graphs

- Topological sort
- Floyd–Warshall algorithm
- Kruskal's algorithm
- Prim's algorithm
- Union–Find (Disjoint Set Union)

Thank you!